GLADYS OUMA

Software Engineer

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EDUCATION

Diploma in Software Engineering

AkiraChix

Feb, 2026 - Nov, 2026

SKILLLS

- **Backend** : Python, FastAPI, REST APIs, PostgreSQL, SQL, API Architecture
- **Data & AI/ML**: Data Analysis, Data Science, Machine Learning, AI Applications, Data Visualization, computer vision, feature engineering
- **Cybersecurity**: STRIDE, access control, data minimization, OWASP Top 10
- **Web Development**: React, Next.js, HTML, CSS, JavaScript
- **Mobile Development**: Flutter, Dart
- **Quality Assurance**: Playwright, Cypress, Postman, API Testing, Functional Testing, Automated Testing
- **Product Management**: Product Requirements, User Stories, Acceptance criteria, Agile/Scrum, Sprint Planning, Jira, Product Roadmap, Prioritization
- **UX**: UX Research, User Interviews, Usability Testing, User Flows, Wireframing, UX/UI Design, Figma, Adobe, Miro,
- **Development Practices**: Git, GitHub, Software Architecture, Documentation, Debugging, Agile Development

REFERENCES

Linda Kamau
Founder & Executive, AkiraChix
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Caroline Mutua
Senior Product Manager, Microsoft
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James Mwai
Co-Founder & CTO Paycloud
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Sharon Obanda
AI Engineering lead, Ona Insights
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Personal Statement

Gladys is passionate about machine learning and exploring data to uncover actionable insights. She has hands-on experience across the full product development lifecycle, working with Python, FastAPI, PostgreSQL, and modern web and mobile technologies. She aspires to grow into roles where she can build reliable, innovative, data-driven systems that create meaningful and lasting impact.

PROJECT EXPERIENCE

Tahadhari- Anti-poaching platform

An offline-first conservation intelligence and patrol prioritization platform for proactive anti-poaching action.

- Designed and implemented the backend architecture using python and FASTAPI to connect the AI prediction engine, commander dashboard, and ranger mobile application into a unified conservation platform.
- Developed the command dashboard monitoring features to give conservation commanders a centralized view of high-risk areas, patrol activities, incidents, ranger assignments, and key operational metrics.
- Implemented functionality that allows commanders to identify and prioritize high-risk patrol areas based on predicted poaching and snaring activity.
- Built the ranger mobile application using Flutter and Dart, enabling rangers to access assignments, record field observations, and update patrol information while working in remote areas.
- Implemented offline-first functionality in the mobile application, allowing rangers to continue accessing essential information when operating in zero-internet environments.
- Trained and integrated an XGBoost machine learning model to analyze conservation data and estimate the probability of poaching or snaring activity occurring in specific areas.
- Prepared and processed historical conservation data for machine learning, transforming relevant environmental, location, patrol, and incident information into features that could be used for risk prediction.
- Implemented risk-based prioritization that helps conservation teams focus limited patrol resources on areas identified as having a higher predicted likelihood of poaching or snaring activity.
- Designed workflows that connect predicted risk information with patrol tasks and field operations, helping commanders move from identifying risks to assigning appropriate patrol areas.
- Created the platform's visual identity, including the Tahadhari logo, dashboard and mobile interface design elements, to establish a consistent product identity.
- Tested the platform across its backend, dashboard, and mobile components using playwright to identify functional issues and verify that key workflows operated correctly across different parts of the system.

Breast Tumor Classification — Computer Vision

An application that analyzes breast ultrasound images and classifies them with the goal of demonstrating how AI can support the identification of potentially cancerous abnormalities.

- Collected, organized and prepared breast ultrasound image data by reviewing image records, grouping images according to their classification, and preparing the dataset for machine learning.
- Preprocessed the images to ensure they were in a consistent format and suitable for model training, including resizing and preparing image data for the CNN.
- Developed and trained a Convolutional Neural Network (CNN) to learn visual patterns in breast ultrasound images and distinguish between benign and malignant cases.
- Configured and evaluated the machine learning model by monitoring its training performance and testing its ability to correctly classify previously unseen images.
- Analysed model performance using classification metrics to identify strengths and weaknesses in the predictions and assess the reliability of the trained model.
- Built an interactive Streamlit application that allows users to upload a breast ultrasound image and receive the model's classification result through a simple web interface.
- Integrated the trained model into the Streamlit application so that predictions could be generated directly from uploaded images rather than requiring users to interact with the underlying machine learning code.